

Alg.test – 2024.05.15. – **A** Neptun code: name:

In this test, you need to write the **name of the applied PoA(s)**, and the **specifications** and draw **small algorithms** on paper. Please look after the **good, readable quality** of your work. There are **80 minutes** available. For a good solution, a **maximum 80 points** could be reached. Don't forget your name and Neptun code! During work you are not allowed to talk (chat), to use cell phones, the internet, books, tutorials, ...

Remember! Cheating is not allowed, and action will be taken.

1: 20p, 2: 20p, 3: 40p

A1. The witches of the Discworld created a database where they store their names, ages, and the number of people cured. Determine whether we have at least one witch who cured more than K people. (Input, output, postcondition, algorithm, PoA's name)

Example input: N=6, K=2005 Esmerelda Weatherwax, 102, 6754 Gytha Ogg, 54, 2341 Magrat Garlick, 100, 1234 Agnes Nitt, 40, 2005 Petulia Gristle, 34, 2012 Letitia de Chumsfanleigh, 20, 1234	Output: yes
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A2. The running time of N car racers was stored for M tests. Give the first car racer's index with the best result (lowest time) for all tests. (Input, output, postcondition, algorithm, PoA's name)

Example input: N=4, M=5						Output: 2
	1. test	2. test	3. test	4. test	5. test	
1. c.racer	4	3	4	4	2	
2. c.racer	9	1	7	3	4	
3. c.racer	2	2	3	9	8	
4. c.racer	3	4	9	3	1	

A3. We have 2 integer sequences. Calculate the sum of the odd numbers that occur only in the first sequence. (Input, output, postcondition, algorithm, PoA's name)

//hint: solve it separately! Try to create functions! First try to define a list from odd numbers of the first sequence only, and then calculate the sum.

Example input: S1: 2, 5, 7, 1, 3, 8, 12 S2: 8, 1, 12, 4, 7	Output: 8 (the sum of: 5, 3)
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Alg.test – 2024.05.15. – **B** Neptun code: name:

In this test, you need to write the **name of the applied PoA(s)**, and the **specifications** and draw **small algorithms** on paper. Please look after the **good, readable quality** of your work. There are **80 minutes** available. For a good solution, a **maximum 80 points** could be reached. Don't forget your name and Neptun code! During work you are not allowed to talk (chat), to use cell phones, the internet, books, tutorials, ...

Remember! Cheating is not allowed, and action will be taken.

1: 20p, 2: 20p, 3: 40p

B1. The witches of the Discworld created a database where they store their names, ages, and the number of people cured. Determine whether we have at least one witch who is older than Y years old. (Input, output, postcondition, algorithm, PoA's name)

Example input: N=6, Y=100 Esmerelda Weatherwax, 102, 6754 Gytha Ogg, 54, 2341 Magrat Garlick, 100, 1234 Agnes Nitt, 40, 2005 Petulia Gristle, 34, 2012 Letitia de Chumsfanleigh, 20, 1234	Output: yes
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B2. The running time of N car racers was stored for M tests. Give the best result (lowest time) for all tests. (Input, output, postcondition, algorithm, PoA's name)

Example input: N=4, M=5						Output: 1
	1. test	2. test	3. test	4. test	5. test	
1. c.racer	4	3	4	4	2	
2. c.racer	9	1	7	3	4	
3. c.racer	2	2	3	9	8	
4. c.racer	3	4	9	3	1	

B3. We have 2 integer sequences. Calculate the sum of the odd numbers that occur in both sequences. (Input, output, postcondition, algorithm, PoA's name)

//hint: solve it separately! Try to create functions! First try to define a list from odd numbers of the first sequence only, and then calculate the sum.

Example input: S1: 2, 5, 7, 1, 3, 8, 12 S2: 8, 1, 12, 4, 7	Output: 8 (the sum of: 7, 1)
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